

2. Software Requirements Specification (SRS) Document

1. Introduction

1.1 Purpose: This document specifies the software requirements for the Restaurant App Minimum Viable Product (MVP), a cross-platform mobile application built using React Native and Expo.

1.2 Scope: The application provides a complete digital dining experience, including user authentication, dynamic menu browsing, cart management with reducers, table reservations, live order tracking, and a dedicated manager dashboard.

1.3 Target Audience: Restaurant customers looking for seamless ordering and table booking, and restaurant managers overseeing daily operations.

2. Overall Description

2.1 Product Perspective: A standalone mobile application operating on React Native with local persistence via AsyncStorage and robust navigation powered by React Navigation (Bottom Tabs & Native Stack).

2.2 User Classes & Characteristics:

Customer: Can browse menus, manage carts, apply promo codes, place orders, reserve tables, and track active orders.

Manager: Has elevated access via the Manager Dashboard to handle incoming orders, review table booking requests, and manage menu availability/pricing

2.3 Operating Environment: Cross-platform mobile devices (iOS/Android via Expo Go) and web browsers.

3. System Features & Functional Requirements

3.1 Authentication & Authorization:

Support for secure login and signup using mock credentials (customer@restaurant.com and manager@restaurant.com with password Password1).

Role-based access control rendering specific tabs (e.g., Manager Dashboard) exclusively for authorized personnel.

3.2 Dynamic Menu & Search:

Categorized menu browsing (Starters, Mains, Desserts, Drinks) with loading states and availability indicators.

Real-time search debouncing (400ms delay) to optimize filtering performance and maintain recent search history chips.

3.3 Cart & State Management (useReducer):

Centralized state handling item additions, quantity increments/decrements, automatic removal at zero quantity, and custom special instructions per item.

Promo code validation (WELCOME10, FEAST20) providing instant percentage-based discounts.

3.4 Order Summary & Performance Optimization:

Computed cost breakdown using useMemo calculating Subtotal, Service Charges (5%), Sales Tax (15%), and Discounts.

Performance optimizations utilizing React.memo and useCallback on list items to minimize unnecessary re-renders.

3.5 Table Reservation Custom Hook:

Custom hook (useReservation) managing table availability checks against party sizes and timeslots.

Form validation ensuring correct Pakistani mobile number formatting (03XX-XXXXXXX).

3.6 Real-time Order Tracking & Manager Dashboard:

Automated order stage progression timer (Pending $\rightarrow$ Preparing $\rightarrow$ Ready $\rightarrow$ Served) paired with an elapsed clock counter.

Manager interface for overseeing order statuses, table bookings, and menu adjustments.

4. Non-Functional Requirements

4.1 Performance: Smooth 60fps scrolling performance through optimized list rendering and debounced search operations.

4.2 Usability: Consistent UI with a dynamic light and dark theme context switcher.

4.3 Reliability: State preservation across sessions using AsyncStorage.